SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I **J. Sai Dheeraj [192311191]** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: J.Sai Dheeraj.

Annexure A

Analytical Problem Solving

[1] RSA Encryption Analysis

Given p=11, q=13

[A] Compute n φ(n)

[B] Choose e=7 and find the private key d

[C] Encrypt the message M=9

[D] Decrypt the ciphertext and verify the result

[E] Analyze how changing e affects security

ANS: Analytical Problem Solving

1. RSA Encryption Analysis

Given:

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

(a) Compute $\boldsymbol{n}$and $\boldsymbol{\phi(n)}$

$$n = p \times q = 11 \times 13 = 143$$
$$\phi(n) = (p-1)(q-1) = 10 \times 12 = 120$$

Answer:

- n = 143
- φ(n) = 120

(b) Find the private key $\boldsymbol{d}$

Find $d$such that

$$7d \equiv 1 \pmod{120}$$

Using the Extended Euclidean Algorithm,

$$d = 103$$

(Check: $7 \times 103 = 721,\ 721 \bmod 120 = 1$)

Private Key:

$$d = 103$$

(c) Encrypt the message

Formula:

$$C = M^e \bmod n$$
$$C = 9^7 \bmod 143$$

Using modular arithmetic,

$$9^7 \bmod 143 = 48$$

Ciphertext:

$$\boxed{C = 48}$$

(d) Decrypt the ciphertext

$$M = C^d \bmod n$$
$$M = 48^{103} \bmod 143$$

Result:

$$\boxed{M = 9}$$

The decrypted message matches the original plaintext.

(e) Effect of changing e

- Smaller e(e.g., 3) → Faster encryption but may introduce security risks if used improperly.
- Larger e→ Slower encryption but generally safer.
- **Common choice:** $e = 65537$because it provides a good balance of security and efficiency.

[2] Block Cipher Mode Evaluation

Given plaintext blocks:

P=[1010,1010,1111,1010], Key = K=1100

[A] Encrypt using ECB and CBC modes (Assume IV = 0000)

[B] Compare the ciphertext outputs

[C] Analyze which mode hides patterns better and explain why

ANS: Block Cipher Mode Evaluation

Given:

Plaintext blocks

P1 = 1010

P2 = 1010

P3 = 1111

P4 = 1010

Key

K = 1100

IV

0000

Assume encryption uses XOR:

$$C = P \oplus K$$

(a) ECB Encryption

Block 1

1010

1100

0110

C1 = **0110**

Block 2

Same plaintext

C2 = **0110**

Block 3

1111

1100

0011

C3 = **0011**

Block 4

Again

C4 = **0110**

ECB Ciphertext

[0110, 0110, 0011, 0110]

CBC Encryption

Formula

$$C_i = (P_i \oplus C_{i-1}) \oplus K$$

IV = 0000

Block 1

$1010 \oplus 0000 = 1010$

$1010 \oplus 1100 = 0110$

C1 = **0110**

Block 2

$1010 \oplus 0110 = 1100$

$1100 \oplus 1100 = 0000$

C2 = **0000**

Block 3

$1111 \oplus 0000 = 1111$

$1111 \oplus 1100 = 0011$

C3 = **0011**

Block 4

$1010 \oplus 0011 = 1001$

$1001 \oplus 1100 = 0101$

C4 = **0101**

CBC Ciphertext

[0110,0000,0011,0101]

(b) Comparison

ECB

0110 0110 0011 0110

CBC

0110 0000 0011 0101

Repeated plaintext blocks produce repeated ciphertext in ECB but not in CBC.

(c) Better mode

CBC is better because:

- Uses an Initialization Vector (IV).
- Each ciphertext block depends on the previous ciphertext block.
- Hides repeating patterns.

[3] DES vs 3-DES vs Blowfish Performance Study

Given:

[A] DES key size = 56 bits, time = 2 ms/block

[B] 3-DES key size = 168 bits, time = 6 ms/block

[C] Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm

ii.Analyze trade-offs between security and performance

iii.Recommend the best algorithm for secure real-time communication

ANS: DES vs 3-DES vs Blowfish Performance Study

Given

Algorithm	**Time/block**
DES	2 ms
3DES	6 ms
Blowfish	1 ms

Blocks = **1000**

(i) Total Encryption Time

DES

$$1000 \times 2 = 2000 \text{ ms} = 2 \text{ s}$$

3DES

$$1000 \times 6 = 6000 \text{ ms} = 6 \text{ s}$$

Blowfish

$$1000 \times 1 = 1000 \text{ ms} = 1 \text{ s}$$

Answer

Algorithm	Total Time
DES	2 s
3DES	6 s
Blowfish	1 s

(ii) Trade-offs

Algorithm	Security	Performance
DES	Low	Medium
3DES	High	Slow
Blowfish	High	Fast

(iii) Recommendation

Blowfish

Reason:

- Fastest encryption
- Strong security
- Suitable for real-time communication

[4] Diffie-Hellman Key Exchange Analysis

Given:

[A] p=23, g=5

[B] User A private key = 6

[C] User B private key = 15

[D] Compute public keys for A and B

[E] Compute the shared secret key

[F] Demonstrate how a Man-in-the-Middle attacker could alter the exchange

[G] Suggest a secure method to prevent the attack

ANS: Diffie-Hellman Key Exchange

Given

- $p = 23$
- $g = 5$
- A private key = 6
- B private key = 15

(d) Public Keys

User A

$$A = 5^6 \bmod 23$$
$$15625 \bmod 23 = 8$$

User B

$$B = 5^{15} \bmod 23$$
$$= 19$$

Public Keys

- A = **8**
- B = **19**

(e) Shared Secret

User A

$$19^6 \mod 23 = 2$$

User B

$$8^{15} \mod 23 = 2$$

Shared secret

$$\boxed{2}$$

(f) Man-in-the-Middle Attack

Attacker intercepts public keys.

Instead of

A → B : 8

attacker sends

12

Instead of

B → A : 19

attacker sends

15

Now

- A shares a key with attacker.
- B shares another key with attacker.
- Attacker decrypts and modifies all messages.

(g) Prevention

Use

- Digital certificates

- Digital signatures
- Authenticated Diffie-Hellman
- TLS

These verify the identity of both users and prevent MITM attacks.

[5] ECC vs RSA Comparison

Given:

[A] RSA key size = 1024 bits

[B] ECC key size = 160 bits (equivalent security)

[C] Device processing capacity = limited (IoT device)

[D] Analyze computational cost for both systems

[E] Estimate which consumes less power and memory

[F] Justify which cryptosystem is better for the given scenario with reasoning

ANS: ECC vs RSA Comparison

Given

- RSA = 1024-bit
- ECC = 160-bit
- IoT device with limited resources

(d) Computational Cost

RSA

- Large keys
- More CPU operations
- Slower

ECC

- Small keys
- Faster calculations
- Lower CPU usage

ECC has lower computational cost.

(e) Power and Memory

Feature	RSA	ECC
Power	High	Low
Memory	High	Low
Key Size	1024-bit	160-bit

ECC consumes less power and memory.

(f) Best Choice

ECC is the better choice because:

- Provides security equivalent to RSA with much smaller keys.
- Requires less computation.
- Consumes less power and memory.
- Is ideal for resource-constrained IoT devices.